

$$\sum_{\alpha, \beta} \rightarrow c\gamma \rightarrow \begin{array}{c} a\alpha \\ | \\ - \\ | \\ b\beta \end{array} + \begin{array}{c} a\alpha \\ | \\ + \\ | \\ b\beta \end{array} \rightarrow c'\gamma' = \rightarrow c\gamma \rightarrow \begin{array}{c} a \\ | \\ - \quad + \\ | \\ b \end{array} \rightarrow c'\gamma'$$

Diagram illustrating a summation over indices α, β and a subsequent simplification of a particle interaction process.

The diagram shows the summation over indices α, β of a process involving an incoming particle $c\gamma$ and two outgoing particles, $a\alpha$ and $b\beta$. This is followed by a plus sign and a similar process where the outgoing particles are $a\alpha$ and $b\beta$, but the internal interaction is different (indicated by the $+$ sign inside the vertex).

The result is shown as an equality to a process involving an incoming particle $c\gamma$ and an outgoing particle $c'\gamma'$, with a loop structure in the intermediate state. The loop consists of two particles, a and b , connected by a vertical line. The top vertex is marked with a minus sign ($-$) and the bottom vertex is marked with a plus sign ($+$).